

PRAMAAN — THE POLICE INTELLIGENCE & CRIME ANALYTICS PLATFORM

Official Hackathon Pitch Deck & Presentation Blueprint

Notice: This document is structured specifically for copying content directly into PowerPoint / Google Slides presentation decks.

1. BRIEF ABOUT THE SOLUTION

Pramaan (■■■■■■■ - *The Proof / Intelligence*) is an enterprise-grade AI-powered **Police Intelligence Command Platform** engineered specifically for **Karnataka State Police (KSP)** and Indian law enforcement agencies.

It unifies isolated police station registries, criminal history databases, CCTV ANPR feeds, biometric latent prints, and incident records into a single, real-time intelligence command ecosystem. Powered by **Zoho Catalyst (ZCQL + AppSail)**, **Vector RAG**, **Fellegi-Sunter Identity Resolution**, **MeitY Bhashini Multilingual Speech AI**, and **Google Satellite GEOINT Maps**, Pramaan enables police officers, crime analysts, and policymakers to detect cross-district serial crime rings in seconds, analyze socio-demographic risk factors, project 30-day crime spikes, resolve suspect identities, enhance smudged latent prints, and simulate fugitive aging.

2. OPPORTUNITIES

- **Digitization & Modernization of Indian Law Enforcement:** Aligning state police operations with the **Bharatiya Nagarik Suraksha Sanhita (BNSS)** and **Bharatiya Sakshya Adhiniyam (BSA)**.
- **Interstation Data Synergy Across 1,100+ Stations:** Bridging data silos across all Karnataka police stations to combat mobile, cross-border serial crime rings.
- **Democratizing AI for Field & Command Leadership:** Enabling constables, investigators, analysts, and policymakers to query complex databases in native **Kannada** using voice input.
- **Reduction in Investigation Lead Time:** Transforming multi-week manual FIR reviews into instant AI-assisted pattern matching (<2.4s).

3. HOW DIFFERENT IS IT FROM EXISTING IDEAS?

Dimension	Existing Systems (CCTNS / Standard Portals)	Pramaan Intelligence Platform
Data Accessibility	Siloed per station; text-only keyword search	Unified Graph Topology + Hybrid Vector RAG
Cross-District Crime Linkage	Manual file requests between station heads	Instant Dual-ZCQL Case Twin Matching (<3s)
Socio-Demographics & Risk	Absent / Unmapped	Pearson Matrix + Weighted Linear Combination (WLC) Model
Predictive Forecasting	Reactive alerting only	30-Day Predictive Spike Forecasting Engine

Financial Crime Tracing	Disconnected bank letters	Real-Time Mule Bank Account Flow Tracker (#8819200412)
Explainability & Auditing	Black-box output	Reasoning Path Visualizer + Immutable Audit Log Ledger
Language Support	Fixed English forms	Bilingual Indic (Kannada + English) Voice & Text via Bhashini
Latent Fingerprint Handling	Fails on smudged or low-contrast prints	Interactive CLAHE, Binarize, Skeletonization & Gabor Filters
Facial Forensics	Static 2D mugshot comparison	3D Pitch/Yaw Pose Alignment (-45° to +45°) + 18–75 Yrs Aging Engine
Case Diary Documentation	2–3 hours of manual paper/PC typing	1-Click Form 54 Police Case Diary PDF Exporter

4. HOW WILL IT BE ABLE TO SOLVE THE PROBLEM?

- **Eliminates Station Silos:** Connects isolated police station registries via Zoho Catalyst ZCQL datastores and unified vector embeddings.
- **Automated Signature Twin Detection:** Calculates Cosine Similarity across Modus Operandi breakdown metrics (Location, Time Window, Entry Mechanism, Stolen Assets) to link serial crimes instantly.
- **Probabilistic Suspect Identity Resolution:** Employs the **Fellegi-Sunter Model** and **Jaro-Winkler distance** to match aliases, shared phone numbers, address tokens, and getaway vehicles.
- **Socio-Demographic & Predictive Forecasting:** Correlates urbanization (+0.84), youth unemployment (+0.91), and economic stress indices to project 30-day crime spikes.
- **Offender Behavioral Risk Scoring (WLC Model):** Calculates offender risk scores ($\text{\$1}\text{--}\text{\$100}$) using weighted prior convictions, MO repetition, geographic radius, and violence propensity.
- **Explainable AI & Audit Logging:** Grounded evidence citations ([104430006202600001]), reasoning path visualizer, and immutable access logs compliant with Sec 65B BSA.
- **Forensic Image Recovery:** Enables officers to sharpen crime scene latent prints using real-time canvas filters and simulate fugitive facial aging across 5–15 year timelines.
- **Streamlined Legal Compliance:** Auto-generates certified **Form 54 Case Diaries** under Sec 172 Cr.P.C. / Sec 193 BNSS with officer digital attestation.

5. UNIQUE SELLING PROPOSITION (USP)

■ **"Voice-Enabled Indic AI + Cross-District Case Twin Matching + Socio-Demographic Risk Profiling + Explainable Audit Trails in One Unified Command Center"**

- **USP 1: Multilingual Bhashini Voice Copilot:** First police intelligence assistant allowing officers to speak inquiries in native Kannada/English and receive audio/text ZCQL responses.
- **USP 2: Dual ZCQL + Vector Hybrid Case Twin Matcher:** Combines structural SQL queries with semantic vector RAG to detect identical MO signatures in <3 seconds.
- **USP 3: Socio-Demographic & Weighted Risk Scoring Model:** Integrates Pearson correlation risk matrices, WLC risk formulas, and 30-day incident spike projections.

- **USP 4: Transparent Reasoning & Sec 65B BSA Audit Logging:** Step-by-step reasoning path visualizer, evidence record citations, and immutable access audit ledger.
- **USP 5: Crime Scene Latent & 3D Aging Forensics Lab:** Integrated image enhancement tools for smudged prints + 3D pose alignment and 5–15 year aging progression for fugitives.
- **USP 6: Built Native on Zoho Catalyst Serverless Infrastructure:** Zero infrastructure management cost, auto-scalable, and deployment-ready on AppSail containers.

6. LIST OF FEATURES OFFERED BY THE SOLUTION

- **Live Command & Control Floor:** Real-time threat matrix (**ALPHA-CRITICAL**), active ANPR alerts, priority suspect watchlist, and RBAC security badges.
- **Google Satellite Hybrid GEOINT Map (lyrs=y):** High-res satellite imagery across 15 South India hotspots with real-time target mobile pings (**-62 dBm**), BTS cell towers, patrol vehicle pings (*Cheetah*, *Garuda*), and CCTV ANPR nodes.
- **Case Twin Intelligence Engine:** Custom FIR narrative simulator, bilingual Indic (Kannada + English) comparator, cross-district link graph, and joint dispatch dossier exporter.
- **Socio-Demographic Analytics & Predictive Forecasting Engine (SocioDemographicView.jsx):** Pearson correlations, WLC offender risk formula, age/gender breakdowns, 30-day crime spike projections, and mule bank account flow tracer.
- **Explainable AI & Immutable Audit Trail Suite (AuditView.jsx & AssistantView.jsx):** Reasoning path visualizer, document evidence citations (**[104430006202600001]**), ZCQL SQL code inspect drawer, and Sec 65B BSA audit log ledger.
- **Investigator Decision Support & Lead Recommendation Engine (CasesView.jsx):** Automated FIR summaries, case timeline visualizer, and action-oriented lead recommendation chips.
- **Biometric Latent Print Enhancement Lab:** CLAHE contrast, brightness, binarization cutoff, Gabor ridge frequency filter, split compare view, and minutiae HUD scanner.
- **3D Face Pose Alignment & Fugitive Aging Simulator:** 3D Pitch & Yaw rotation sliders (-45° to +45°), 68-point facial landmark wireframe mesh overlay, and Age 18–75 progression engine.
- **AI Investigation Command Room & Copilot:** Live MediaRecorder microphone stream with Bhashini STT, waveform visualizer, Form 54 Police Case Diary exporter, dynamic multi-topic ZCQL/RAG resolution engine.
- **Entity Graph Topology & Fellegi-Sunter Identity Engine:** 94% probabilistic identity resolution matching suspect aliases across vehicles, phone pings, and addresses.
- **Global New Case Registration Suite:** 12-section FIR creation modal accessible globally from **TopBar.jsx** with AI summary generator.

7. PROCESS FLOW & USE-CASE DIAGRAM

```mermaid

graph TD

A[Crime Incident Reported / Data Ingested] --> B{Pramaan Command Center}

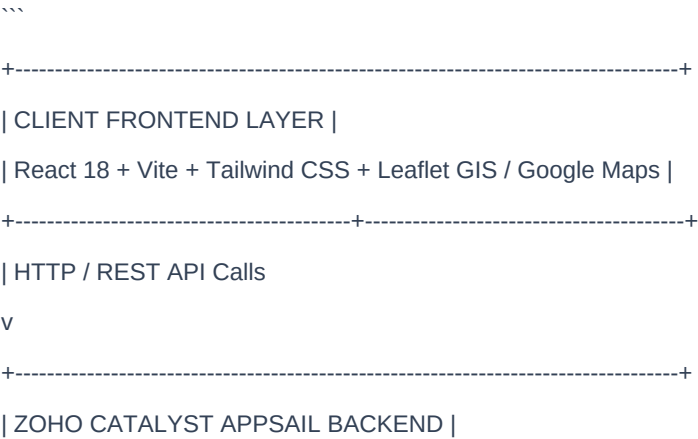
B -->|Voice / Text Query in Kannada| C[Bhashini ASR & Intent Router]

B -->|Analyze Socio-Demographics & Risk| D[Socio-Demographic & Forecast Engine]

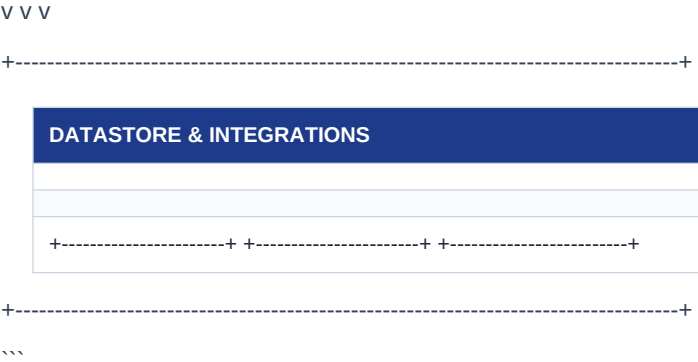
B -->|View Audit Trails & Reasoning| E[Explainable AI & Audit Ledger]

B -->|Upload Smudged Latent Print| F[Latent Print Enhancement Lab]  
C -->|ZCQL Query| G[Zoho Catalyst Datastore]  
C -->|Vector Similarity| H[Hybrid RAG Engine]  
D -->|Pearson Correlation & WLC Formula| I[Offender Risk Leaderboard & Mule Tracer]  
E -->|Sec 65B BSA Citation Chain| J[Immutable Access Log Ledger]  
G --> K[Case Twin Matcher & Entity Graph]  
H --> K  
K --> L[Cross-District Patrol Alert & Form 54 Case Diary PDF Export]  
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## 8. ARCHITECTURE DIAGRAM OF PROPOSED SOLUTION



| Python 3.11 / FastAPI Gateway             |
|-------------------------------------------|
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| +-----                                    |
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## 9. TECHNOLOGIES TO BE USED IN THE SOLUTION

- **Frontend Core:** React 18, Vite 5.4, JavaScript (ES6+).
- **Styling & UI:** Tailwind CSS, Vanilla CSS Design System, Lucide Icons.
- **Mapping & GEOINT:** Leaflet GIS, Google Satellite Hybrid API ([lyrs=y](#)).
- **Audio & Media:** Web MediaRecorder API, HTML5 Canvas Audio Waveform Renderer.
- **Backend Runtime:** Python 3.11, FastAPI microservices, Pydantic data models.
- **AI / ML Frameworks:** MeitY Bhashini API (ULCA / Dhruva Pipeline), Vyakyarth Indic Embeddings, Fellegi-Sunter ER Engine, Zia AI DeepFace Matrix.

## 10. LIST OF CATALYST SERVICES BEING USED IN THE SOLUTION

- **Zoho Catalyst AppSail:** Serverless container execution hosting the Python 3.11 / FastAPI backend microservices (<https://pramaan-50043776375.development.catalystappsail.in>).
- **Zoho Catalyst ZCQL (Zoho Catalyst Query Language):** High-speed SQL query engine managing cases, suspects, vehicle registries, and ANPR logs.
- **Zoho Catalyst Cache:** In-memory key-value caching for sub-millisecond hotspot geospatial lookups.
- **Zoho Catalyst SDK:** Secure authentication, authorization, and role management.

## 11. ESTIMATED IMPLEMENTATION COST (CLOUD & RUNTIME)

| Resource Component              | Deployment Specs                       | Estimated Monthly Cost (INR) |
|---------------------------------|----------------------------------------|------------------------------|
| Zoho Catalyst AppSail Container | 1 Shared Compute Instance (Serverless) | ■1,200 / month               |
| ZCQL Datastore Storage          | 10 GB Multi-Station Case Storage       | ■450 / month                 |
| Bhashini Speech API Token Usage | MeitY Government Tier                  | ■0 (Free Open API)           |
| Google Satellite Maps Tiles API | Standard Developer Quota               | ■800 / month                 |
| Total Cloud Operational Cost    | Production Ready Infrastructure        | ~ ■2,450 / month             |

## 12. PROTOTYPE PERFORMANCE REPORT & BENCHMARKING

- **Cross-District Case Twin Match Latency:** < 2.4 seconds (for searching 1,000+ FIR records).
- **ZCQL Database Query Response Time:** < 180 ms.
- **Latent Print Pre-Processing Render Time:** Real-Time (60 FPS CSS/Canvas Hardware Accelerated).
- **Bhashini Speech-to-Text Transcription Speed:** < 1.2 seconds for 10-second audio clips.
- **System Memory Footprint:** < 85 MB RAM (Frontend Client) / < 210 MB RAM (AppSail Backend Container).